

Deconver for Stroke Lesion Segmentation

Nonnegative deconvolution instead of self-attention: best Dice of four architectures on ISLES’26 at a third of the parameters

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63.6%

Best mean Dice of nnU-Net, SegResNet, SwinUNETR-V2 and Deconver, five-fold cross-validation on ISLES’26

3×

Fewer parameters than either large baseline: 25.5M against 75.2M (SegResNet) and 72.8M (SwinUNETR-V2)

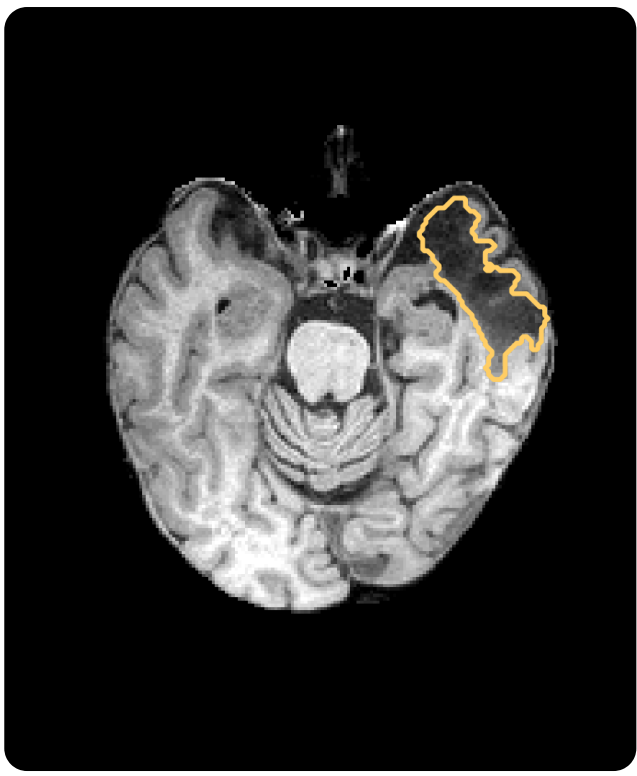
64.4%

Dice of the SegResNet + Deconver ensemble, our challenge submission: the two blocks err differently

01 The problem ISLES’26

Segment ischemic stroke lesions from T1-weighted MRI alone, across lesion stages and imaging centers.

- **No diffusion contrast.** The DWI/ADC signal that acute delineation usually relies on is absent.
- **Acute to chronic lesions**, whose T1 appearance changes markedly with time since onset.
- **More than 60 centers** at native resolution: spacing, field of view and orientation all vary.
- Generalization is the operative difficulty, not accuracy under harmonized acquisition.



Native-space T1w scan with the expert lesion mask (yellow)

Which network block copes best? Published comparisons confound the block with pipeline engineering and model size. We fix everything except the block.

02 Key idea Deconver

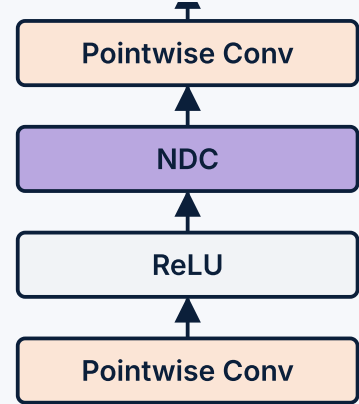
Keep the U-shaped backbone and the transformer block layout, but replace self-attention with a **Deconv Mixer** built around a nonnegative deconvolution (NDC) layer.

DECONVER BLOCK

$$\mathcal{Z} = \text{DeconvMixer}(\text{IN}(\mathcal{X})) + \mathcal{X}, \quad \mathcal{Y} = \text{MLP}(\text{IN}(\mathcal{Z})) + \mathcal{Z}$$

Instance norm (IN) instead of layer norm suits small 3D batches; the MLP is two pointwise convolutions with a GELU.

DECONV MIXER



Project, enforce nonnegativity, deconvolve, project back

NDC LAYER: ONE RICHARDSON-LUCY STEP

The nonnegative feature map X is treated as a **blurred observation** of a latent source S , seen through a learnable nonnegative filter V :

$$\min_{S \geq 0} \|X - S * V\|_F^2, \quad V \geq 0$$

From a learned initialization $S^{(0)}$, a single multiplicative update recovers a sharper source (V^- : adjoint filter, $\epsilon = 10^{-8}$):

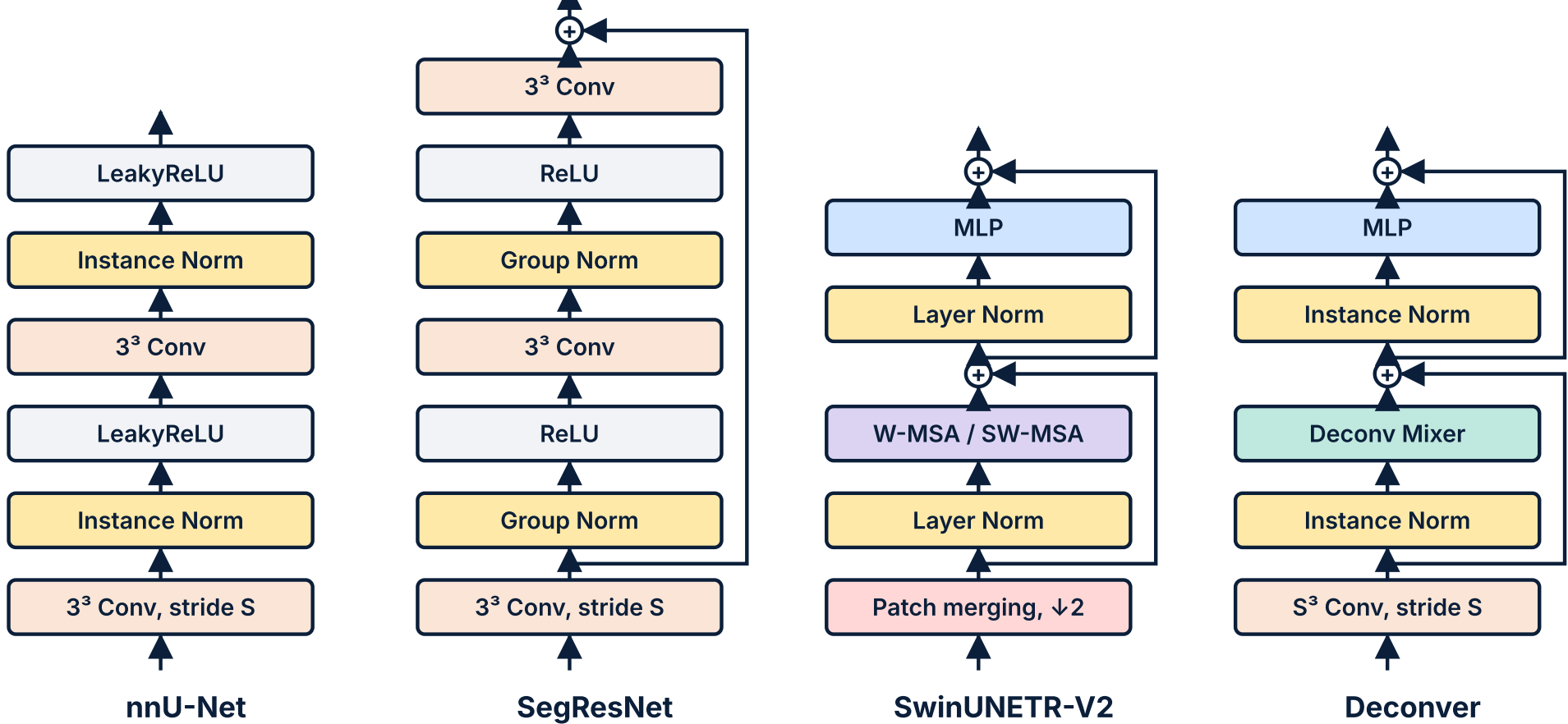
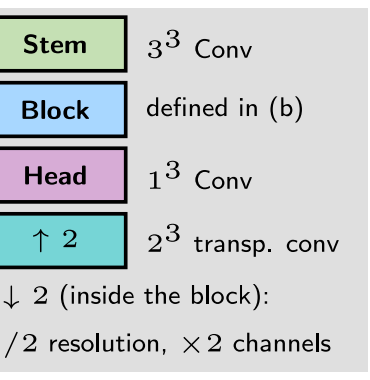
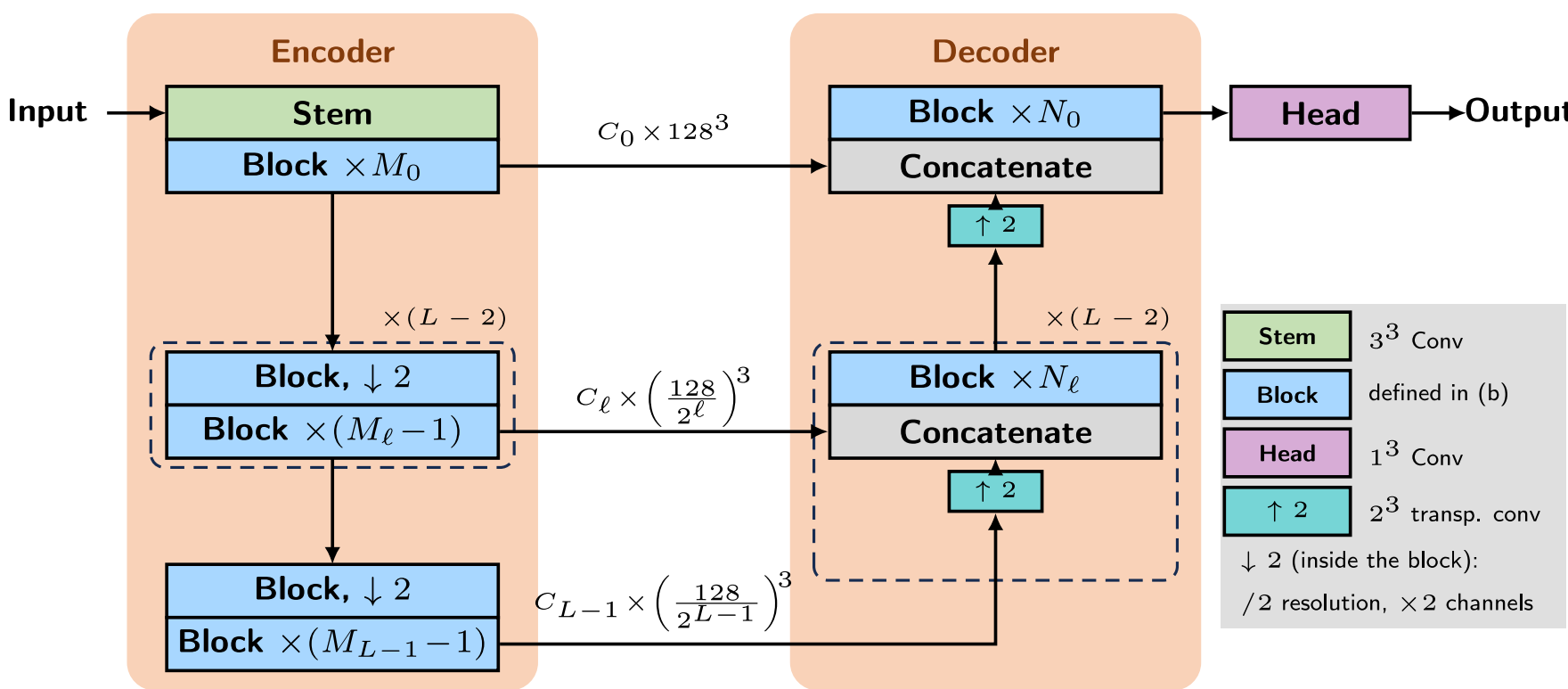
$$S^{(1)} = S^{(0)} \odot \frac{X * V^- + \epsilon}{(S^{(0)} * V) * V^- + \epsilon}$$

- ✓ **Provably safe:** never increases the reconstruction error, keeps nonnegativity, is differentiable: V is trained end to end.

- ✓ **Cheap context:** depthwise groups, expansion ratio 4, one iteration: spatial dependencies without attention's cost.
- ✓ **Restores high-frequency detail** while suppressing artifacts; descends from Factorizer (3rd in ISLES’22) and keeps its nonnegativity prior.

03 Controlled comparison one pipeline, four blocks

All four networks share the encoder–decoder backbone of Fig. 1a: level ℓ carries C_ℓ channels and holds M_ℓ encoder and N_ℓ decoder blocks; each stage halves the resolution and doubles the channels. **Only the block repeated at each level differs** (Fig. 1b; data flows upwards, $S = 2$ in the first block of a stage).



PREPROCESSING

Foreground crop, RAS, 1×1×1 mm, z-score of non-zero voxels

VALIDATION

Five folds stratified by lesion volume, identical for all models (1,451 cases)

TRAINING

128³ patches, batch 2, AdamW 10⁻⁴, warm-up cosine, 300 epochs, Dice + CE

INFERENCE

Sliding window, 128³, mean Dice over the validation cases of all folds

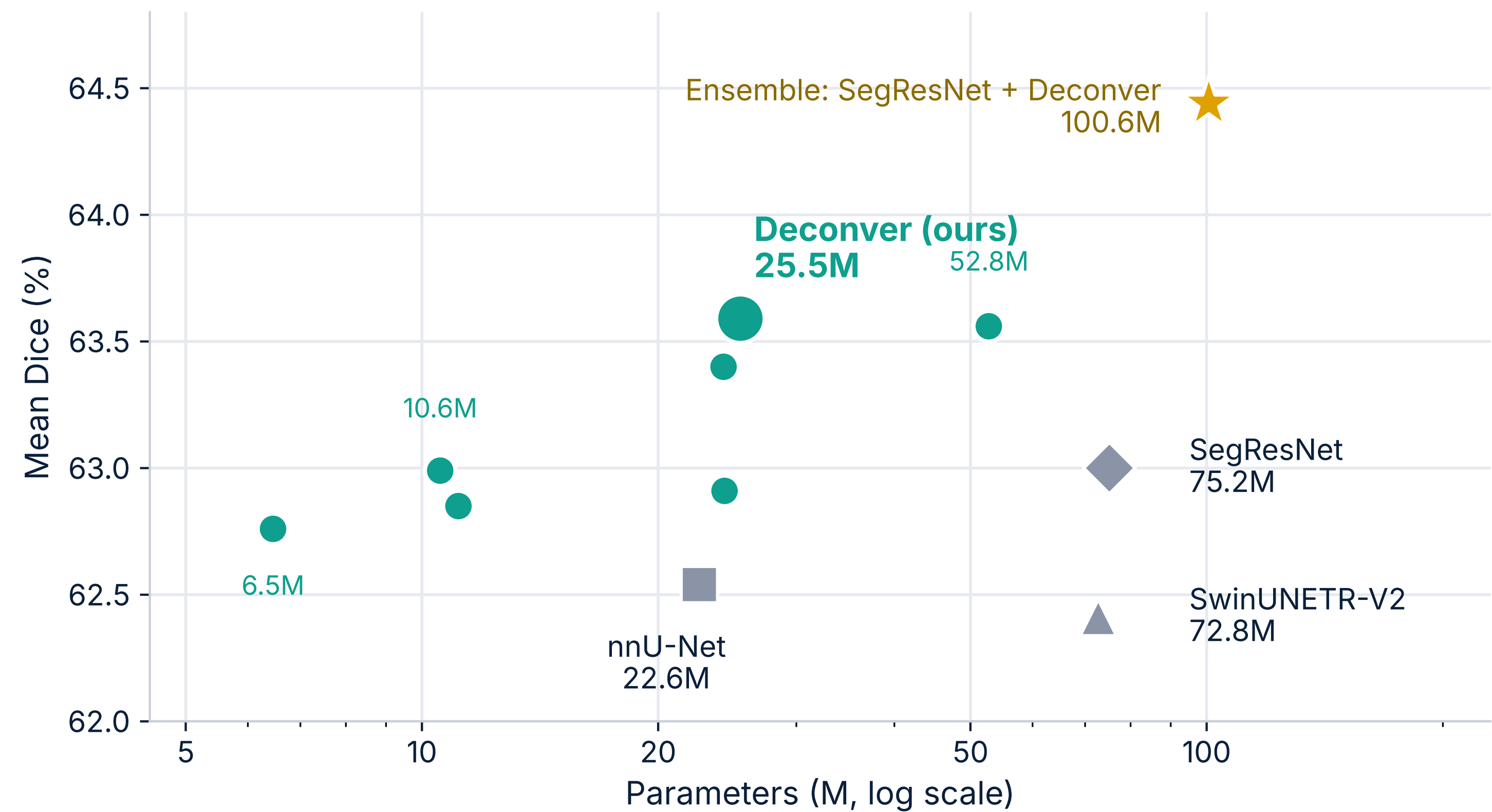
AUGMENTATION

Affine, Gaussian noise and blur, intensity scale and shift (p = 0.2), flips (p = 0.5)

ENSEMBLE

Voxelwise mean of the SegResNet and Deconver probability maps (10 networks)

04 Results accuracy is not aligned with capacity

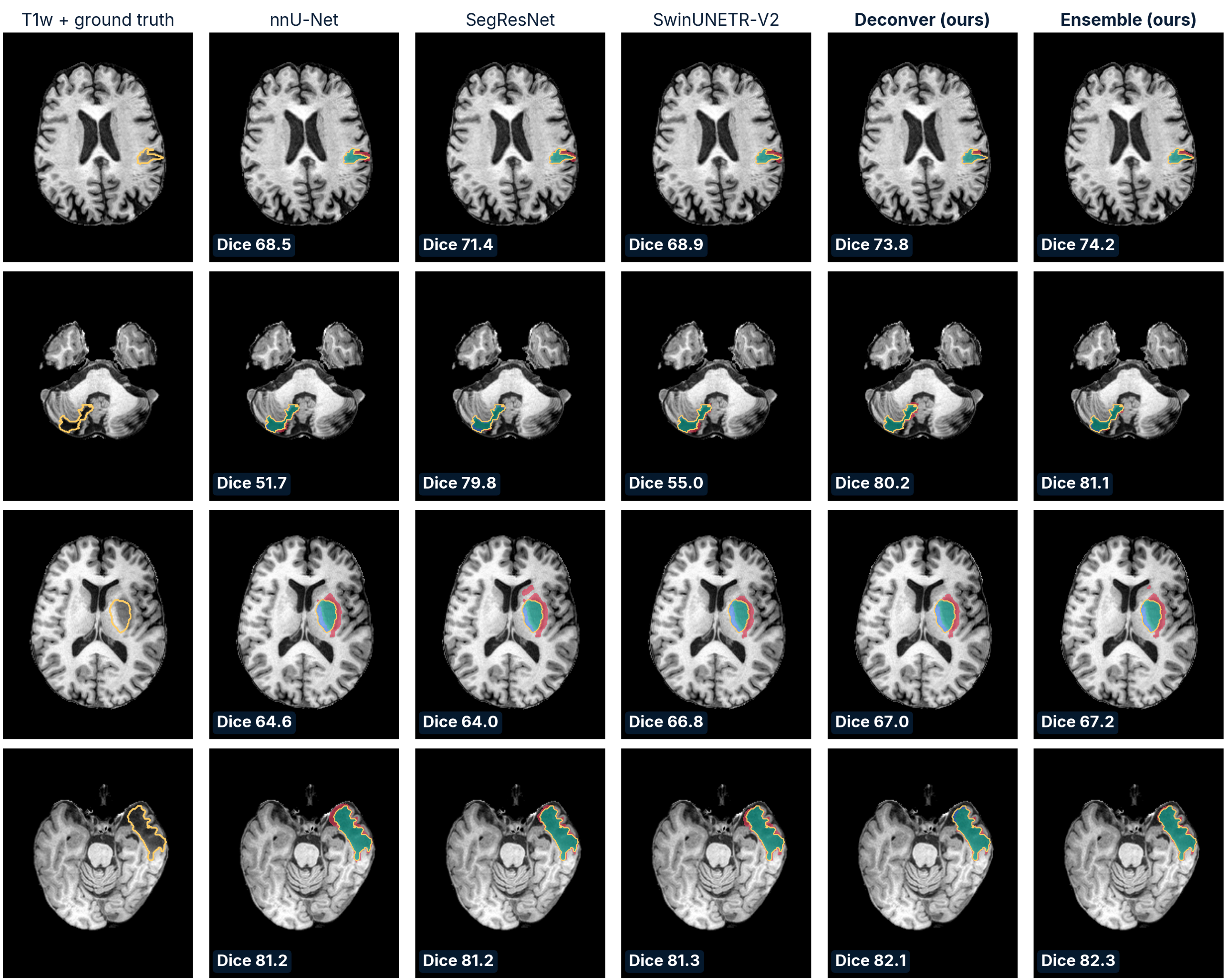


Teal: the seven Deconver configurations of the paper’s ablation (large dot: the model of Table 1). Gray: baselines. Star: SegResNet + Deconver ensemble.

MODEL	$[M_\ell]$	C_0	PARAMS (M)	DICE (%)
nnU-Net	[1,1,1,1,1]	32	22.57	62.54
SegResNet	[1,2,2,4]	64	75.17	63.00
SwinUNETR-V2	[2,2,2,2]	48	72.76	62.41
Deconver	[1,2,2,4]	64	25.46	63.59
Ensemble (SegResNet + Deconver)			100.63	64.44

+0.59 Dice over the strongest baseline at 2.95× fewer parameters; +1.18 over SwinUNETR-V2 at 2.9× fewer. nnU-Net, the only baseline of comparable size, trails by 1.05.

05 Qualitative results unseen validation cases



■ true positive ■ false positive ■ missed lesion ■ ground-truth contour

Four fold-0 validation cases (3 to 52 mL lesions, never seen in training) segmented by the fold-0 models of each architecture and by their SegResNet + Deconver ensemble, with the Dice of each case. In every row the ensemble scores highest, followed by Deconver, ahead of all three baselines, as in the cross-validation.

ABLATION: WIDTH HELPS, DEPTH AND KERNEL SIZE DO NOT (Table 2 of the paper)

$[M_\ell]$	C_0	K	PARAMS (M)	DICE (%)	TAKEAWAY
[1,2,2,4]	32	3	6.46	62.76	smallest variant, still beats nnU-Net and SwinUNETR-V2
[1,1,1,1,1]	32	3	10.55	62.99	matches SegResNet at 7.1× fewer parameters
[1,1,1,1,1]	32	5	11.13	62.85	larger NDC kernel: no gain
[1,1,1,1,1]	64	3	24.23	63.40	twice the width: +0.41
[1,1,1,1,1,1]	32	3	24.30	62.91	sixth level: 2.3× the parameters, no gain
[1,2,2,4]	64	3	25.46	63.59	twice the width: +0.83; the model of Table 1
[1,2,2,4,4]	64	3	52.77	63.56	fifth level: 2× the parameters, no gain

06 Take-home messages

- **Nonnegative deconvolution is a strong block for stroke lesions:** under one fixed pipeline, Deconver gives the best mean Dice of the four architectures (63.59) at about a third of the parameters of the two large baselines.
- **Scale the width, not the depth or the kernel:** all seven Deconver variants span only 0.83 Dice across an eightfold parameter range.
- **Combining blocks beats enlarging them:** averaging SegResNet and Deconver probability maps adds +0.85 Dice (64.44), more than any scaling step; this ten-network ensemble is our challenge submission. **Next:** lesion-wise metrics, per-fold tests, center-held-out splits.

Try Deconver

github.com/pashtari/deconver

`pip install git+https://github.com/pashtari/deconver.git`

ISLES’26 configs, trained models and results:

github.com/pashtari/deconver-isles26

Vyncke, Noei, Jurman, Pižurica, Ashtari. *Deconver for Stroke Lesion Segmentation*. SWITCH+ Workshop, MICCAI 2026. Ashtari et al. *Deconver: A Deconvolutional Network for Medical Image Segmentation*. IEEE JBHI, 2025.

